CSE 3311- ITERATION 2 DOCUMENT

MAVWALK DESIGN DOCUMENT

TEAM 9 MEMBERS: Cleona Hua, Saina Shrestha, Paul Dang, Jaafar Alumary

Repository:

https://github.com/ceh6514/MavWalk

Background Context

In the first iteration, MavWalk was planned as a campus safety escort app in partnership with UTA Police. However, after communicating with UTA Police and Parking & Transportation, we learned that direct integration wasn't possible due to limited staffing; only two officers are available during night shifts and internal system restrictions. Because of this, our team decided to pivot the project to something more practical and achievable within our scope.

The new MavWalk focuses on helping students navigate campus through UTA-specific walking routes that general map apps don't show, like shortcuts through buildings or safer, well-lit areas. To maintain the spirit of connection from the original idea, the app also includes anonymous motivational messages written by Maverick Fellows to create a sense of encouragement and positivity for students walking across campus.

Problem Context

While UTA is a vibrant campus, many students still experience moments of stress or isolation, especially when walking alone between classes or dorms. Existing map apps provide directions but lack the personal or emotional touch that makes students feel connected. At the same time, social media often adds pressure instead of genuine encouragement.

MavWalk aims to bridge that gap by combining simple navigation with positive, uplifting messages, turning everyday walks into small moments of connection and motivation.

Goals

The primary goals of the MavWalk project are to:

- **Combat Loneliness:** Create a sense of passive, positive community by allowing students to receive anonymous, kind messages as they navigate the campus.
- **Provide Delightful Navigation:** Offer a simple, reliable web-based tool for walking directions along a curated set of safe, pre-defined routes at UTA.
- **Foster a Positive Campus Culture:** Encourage students to "pay it forward" by leaving their own kind messages for others after completing a walk.
- **Achieve a High-Quality MVP:** Develop a focused, low-risk Minimum Viable Product by eliminating complex external dependencies , live tracking, and the initial need for content moderation.

Non-Goals

To ensure the project remains focused and achievable, the following are explicitly out of scope for the MVP:

- **Emergency Services:** This is not a safety or emergency application. It will not include an S.O.S. button or any features that imply a connection to emergency services, thereby avoiding the liability and risk associated with such features.
- **User Accounts:** The application will not require user registration or logins. This simplifies development, enhances user privacy, and encourages participation without barriers.
- **Dynamic Route Generation:** The MVP will not generate routes on the fly. All routes will be pre-defined to ensure they are safe, accurate, and efficient for the UTA campus environment.
- **Live User-Generated Content:** To eliminate the significant risk of content moderation at launch, the application will initially only display a curated set of pre-generated, positive messages. The feature for users to submit their own messages will be built, but these messages will be saved for future review and potential use post-MVP.

Proposed Solution

MavWalk will be a responsive web application designed with a mobile-first approach, ensuring accessibility for all UTA students on any device with a web browser. The core experience is simple: a student selects their starting point and destination from a pre-defined list of campus locations. Before the route map is displayed, the user is presented with a full-screen, anonymous, kind message. After viewing the message, they are shown a map with their static walking path.

Upon completing the route, they are given the option to anonymously write and submit their own kind message to be left along that path for a future traveler.

This solution is advantageous because it is technically simple, has no external stakeholder dependencies, respects user privacy by avoiding live tracking, and is centered on a clear, positive mission.

Quantifiable Metrics

Requirement	**Measurement**	**Target**
Route loading time	Response time	≤ 2 s
Predefined routes	Count	10
Curated messages	Count	50
Beta user satisfaction	Positive responses	≥ 85 %
Accessibility (Lighthouse)	Score	≥ 90 %

Testability Plan

Requirement	**Test Type**	**Expected Outcome**
Route selection and display	Functional	Correct route shown for all options
Message display	UI Test	Random message appears before map view

Data privacy	Security	Unapproved messages never displayed
Load performance	Performance	App loads in under two seconds
Accessibility	Usability	Passes mobile readability and contrast tests

Stakeholders & Collaborations

- **UTA Campus Police:** Provided insight into existing escort operations. Due to staffing limits (only two officers available during nighttime shifts) and internal system access restrictions, direct technical integration was deemed infeasible for the MVP.

- **UTA Parking & Transportation:** Contacted to discuss integration with the MavMover app; no formal response received. This informed the decision to pivot to a self-contained project.

- **UTA IT Department:** Consulted informally for potential system access. They confirmed integration would require official campus authorization and could not be supported within the class project's timeline.

- **UTA Students:** Identified as primary users. Feedback will be collected during the closed beta to evaluate usability and engagement.

MoSCoW Prioritization for Iteration 2

Must Have: Route selection, kind-message screen, static map, message submission, responsive design

Should Have: Route-completion confirmation, database integration, curated message storage

Could Have: Route duration estimate, daily quote feature

Will Not Have: SOS button, live tracking, login system, real-time content moderation

Design

System Flow Diagram:
[User Selects Route from Dropdowns] →
[App Fetches & Displays a Random Kind Message] →
[User Continues to Map Screen] →
[App Displays Static Route on Map] →
[User Completes Walk] →
[App Presents Option to Leave a Message]

Screen Flow:

1. **Home Screen:** The main page will feature the MavWalk logo and two dropdown menus: "Starting From" and "Destination." A single "Find My Route" button initiates the journey.
2. **Message Screen:** After a route is selected, this screen appears, displaying one pre-generated kind message. A "Continue" button takes the user to the map.
3. **Map Screen:** This screen displays a map (via React Leaflet) with the pre-defined route drawn as a clear path. An "I've Arrived" button is located at the bottom.
4. **Completion Screen:** Upon finishing, this screen confirms "Route Completed!" and presents an optional text box where a user can type a message. A "Send Message" button submits their contribution, and a "Finish" button returns them to the Home Screen. To be noted that the user can just press the “Finish” button without hitting “Send Message.”

UI Distinct Elements:

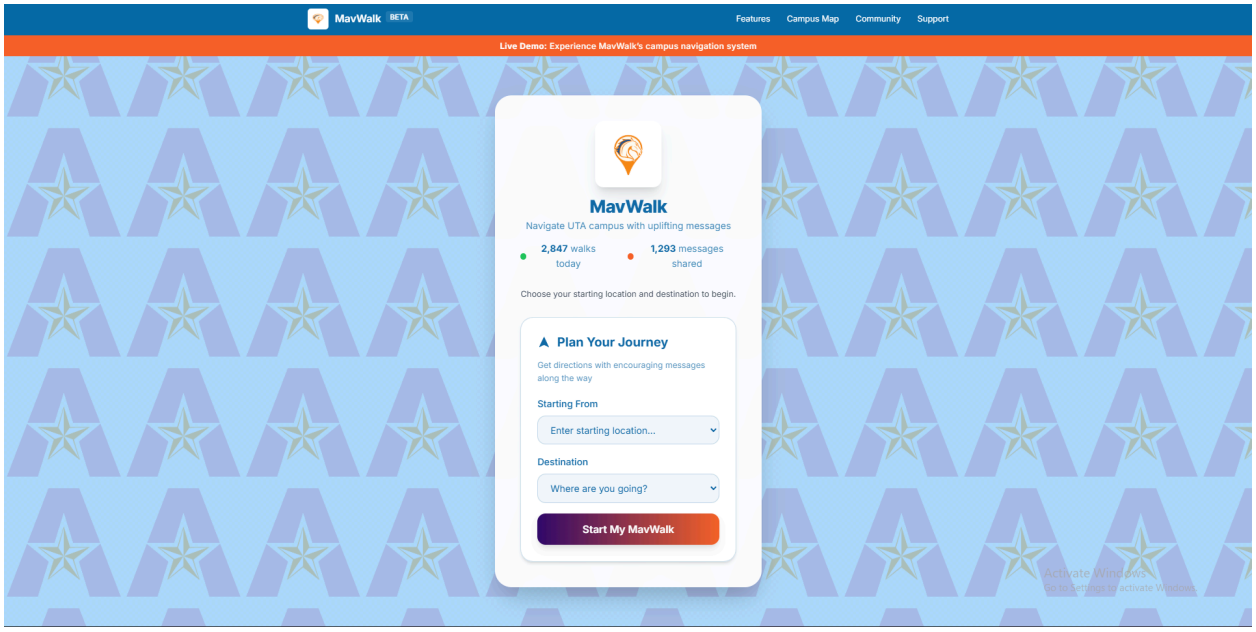
To improve visual clarity and meet accessibility standards, the ETA (Estimated Time of Arrival) on the map screen will be displayed in bold text and a contrasting color (such as blue or orange) to distinguish it from other map labels and ensure readability. Additionally, the MavWalk logo will be featured prominently on the Home Screen to establish clear branding and visual identity for the app. These enhancements address prior feedback about text visibility and missing visual elements.

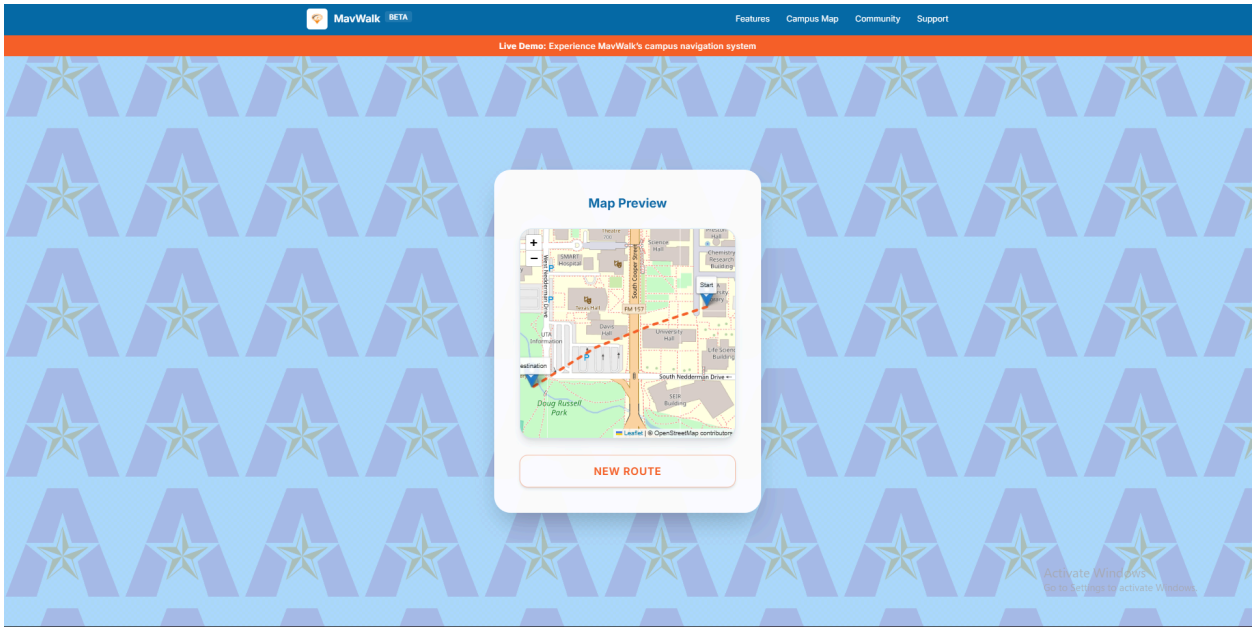
Data Model: The backend database will consist of two primary tables:

- Routes: Contains route_id, start_location_name, end_location_name, and route_path_coordinates (a stored array of latitude/longitude points).
- Messages: Contains message_id, message_text, associated_route_id (optional), and a boolean is_approved flag for future moderation.

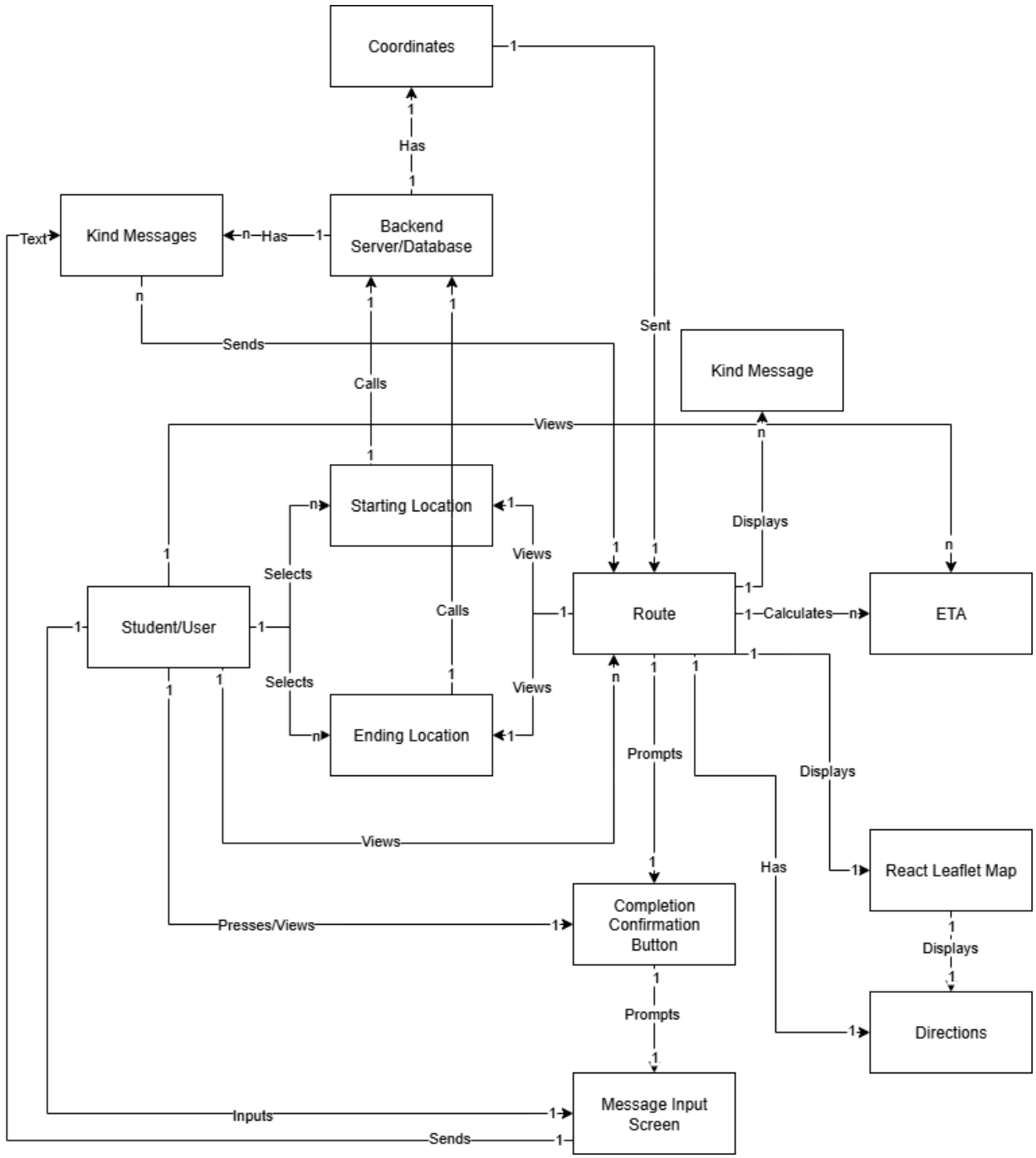
User Interface Design (Iteration 2 UI)

(Zoomed out to 80% to fit in one screenshot)





System Design



Alternatives Considered

During the initial design phase, we evaluated several existing navigation and safety solutions to identify what gaps MavWalk could fill for UTA students:

- Google Maps: Provides accurate global navigation but lacks campus-specific details like indoor building shortcuts, sidewalks, and lighting conditions. It also does not promote social connection or positive interaction among students.

- Apple Maps: Offers basic walking routes but performs inconsistently on campus due to limited pedestrian data and less frequent updates for local environments. It similarly lacks community engagement features.

- UTA MavMover App: Focuses on bus transportation around campus rather than pedestrian routes. It does not support walking navigation or allow students to interact or leave feedback.

- Original "UTA Police Escort" Concept: Initially proposed as a live-tracking safety app, this idea was pivoted after discovering integration and staffing limitations within UTA Police and Parking & Transportation.

These comparisons showed that while traditional map apps excel at navigation, they do not address emotional well-being or campus-specific walking experiences. MavWalk aims to fill that gap by combining curated walking routes with uplifting, community-driven messaging.

Risk Exposure (RE) Analysis

Risk	**Risk Description**	**Category**	**P(0-1)**	**E(Impact)**	**RE**	**Why it matters**	**Mitigation**	**Contingency**
R1	Wrong functionality / misaligned with student needs	Business	0.5	100	50.0	Main risk identified in slides, building features without user feedback	Early student feedback via beta test and QR survey	If feedback negative, refine routes and message concepts
R2	Low adoption / engagement	Business	0.7	40	28.0	App depends on students actually using it	Simplify UI; no login; promote via dorms	Refocus on single popular route first or engage in lots of on campus

							and library posters	advertisement
R3	Technical bug / map integration failure	Technical	0.6	20	12.0	Third-party API or map rendering failure	Use static JSON routes and offline testing	Fallback to screenshot map display
R4	Accessibility / usability failures	Quality	0.3	40	12.0	Affects inclusivity and grading	Mobile-first design; test WCAG AA	Increase font/contrast if needed
R5	Team member availability / burnout	Resource	0.9	8	7.2	Everyone has other classes and jobs	Shared calendar and light PR cycle	Redistribute workload
R6	Performance issues (> 2s load)	Technical	0.5	10	5.0	Slow load time affects user satisfaction	Cache routes; compress assets	Fallback to static map
R7	Inappropriate content (submitted messages)	Content/ Security	0.5	8	4.0	Could violate UTA guidelines or harm reputation	Use only curated messages for MVP	Disable 'submit message' feature and update filtering algorithm

Test Cases

TC-01: Successful End-to-End Route Completion ("Happy Path")

- **Objective:** To verify that a user can successfully select a route, view a message, see the map, and complete the journey as intended.
- **Test Steps:**

    1. On the Home Screen, select a starting point (e.g., "Library") from the "Starting From" dropdown.
    2. Select a destination (e.g., "Science & Engineering Innovation & Research Building") from the "Destination" dropdown.
    3. Click the "Find My Route" button.
    4. On the Message Screen, click the "Continue" button.
    5. On the Map Screen, observe the route and click the "I've Arrived" button.
    6. On the Completion Screen, type "Have a great day!" into the text box.
    7. Click the "Send Message" button, then the "Finish" button.
- **Expected Result:** The user is seamlessly guided through all four screens without errors. After clicking "Finish," the user should be returned to the Home Screen. The submitted message should be saved to the database with its `is_approved` flag set to `false`.

TC-02: Verify Unapproved Messages Are Not Displayed

- **Objective:** To ensure that user-submitted messages are saved for future review and are not immediately shown to other users, protecting data privacy and preventing the display of unmoderated content.
- **Test Steps:**
    1. Successfully complete a route and submit a unique, identifiable message (e.g., "Testing123-Purple-Elephant").
    2. Return to the Home Screen.
    3. Begin the *same route* again.
    4. Repeat this process 10 times to cycle through multiple messages.
- **Expected Result:** The unique message "Testing123-Purple-Elephant" should **never** appear on the Message Screen. Only the pre-generated, curated messages from the database should be displayed.

TC-03: Complete a Route Without Submitting a Message

- **Objective:** To confirm that submitting a message is optional and that a user can finish a route without any issues after choosing not to write one.
- **Test Steps:**
    1. Complete all steps for a route until you reach the Completion Screen.
    2. Do not type anything into the message text box.
    3. Click the "Finish" button directly, without clicking "Send Message".

- **Expected Result:** The application should immediately return to the Home Screen without any errors. No new message entry should be created in the database.

TC-04: UI Element Verification for Accessibility

- **Objective:** To verify that specific UI elements meet the design and accessibility requirements outlined in the document.
- **Test Steps:**
    1. Select any route to navigate to the Map Screen.
    2. Inspect the "ETA (Estimated Time of Arrival)" text on the map.
- **Expected Result:** The ETA text must be **bold** and displayed in a **contrasting color** (like blue or orange) to ensure it is clearly distinguishable and readable, as specified in the UI Distinct Elements section.

TC-05: Performance and Load Time Measurement

- **Objective:** To ensure the application meets its target performance metric for route loading time.
- **Test Steps:**
    1. Using a browser's developer tools, prepare to measure the load time.
    2. On the Home Screen, select a starting point and destination.
    3. Click the "Find My Route" button and start the timer.
    4. Stop the timer as soon as the Message Screen is fully rendered and displayed to the user.
- **Expected Result:** The total response time from clicking the button to the Message Screen appearing should be **less than or equal to 2 seconds** (≤ 2 s).

User Feedback and Validation Plan:

To gather qualitative feedback, the 5 UTA students from on-campus residence halls are utilizing our website locally by cloning it to their computers from github. We will try to get our website on google cloud services in order for it to be used on more devices.

Participants will be asked to “complete” three specific routes and submit one message. Afterward, they will be sent a short Google Form survey with questions targeting:

- Usability: How easy was it to select and follow a route?
- Engagement: Did the kind messages positively impact your “walk?”
- Clarity: Were there any confusing screens or buttons? We are aiming for an 80% positive response rate on overall satisfaction to validate our MVP.

Open Questions

The focused MVP plan has resolved most open questions. The primary remaining question for future iterations is:

- What is the most effective and scalable content moderation strategy (e.g., community voting, AI filtering, manual review) to implement when enabling user-submitted messages to be displayed publicly?

Parties Involved

This project will be developed by Team 9.

- **Team Members:** Cleona Hua, Saina Shrestha, Paul Dang, Jaafar Alumary
- **Roles:** Paul Dang will lead front-end development, and Cleona Hua will lead back-end development. Saina Shrestha will lead the UI/UX design and documentation, creating wireframes and visual mockups to guide development and maintain project documentation for each iteration. Jaafar will lead the presentation slide and documentation.

Timeline and Milestones

- **Iteration 2: Refactoring, Foundation, & Content (2 Weeks):**
    - Revise the Requirements sheet
    - Design documents overhaul
    - Set up backend service and database
    - Build the React front-end, including all four primary screens.
    - Integrate React Leaflet to display the static route paths from the database.
    - Acquire user feedback on MVP features
- **Iteration 3: Integration & Testing (3 Weeks):**
    - Connect the front-end to the backend API to fetch routes and messages.

    - Implement the logic for submitting new messages to the database (to be stored for later review).
    - Define and populate the Routes table with 10 core campus routes.
    - Define and populate the Messages table with 50 approved, pre-generated messages.
    - Conduct thorough internal team testing of the full user flow.
- **Final Deliverable: Beta Launch & Feedback (Ongoing):**
    - Deploy the web application to a live URL using google cloud services.
    - Recruit a small group of on-campus residents for a closed beta test and gather feedback for future improvements.
    - Continuous updating with different routes
    - Further testing

Summary

The revised MavWalk project focuses on a simple, high-impact MVP that combines functional navigation with positive social interaction. This iteration addresses all feedback by adding quantifiable requirements, user feedback mechanisms, data structures, and testing plans. It provides a clear, testable roadmap toward a stable and meaningful product for UTA students.

References

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